AURORA K

HUE

WORKING TITLE · WHITEPAPER V0.1

A private palette protocol for beauty and colour: owned by her, computed in hardware, settled on Solana.



Confidential draft · Prepared for the MagicBlock Solana Blitz hackathon · July 2026
Built with MagicBlock Private Ephemeral Rollups · Designed for the Solana Mobile Seeker

CONTENTS

What this paper covers

This is a v0.1 draft. It exists to make one argument clearly, then show that the argument can be shipped in a weekend.

01 **The problem** — *beauty tech fails the women it studies least*

02 **The idea** — *ownership flips the data problem*

03 **The experience** — *what she sees, taps and earns*

04 **Architecture** — *device, agent, enclave, chain*

05 **Privacy and security** — *the threat model, answered*

06 **Why Solana, why MagicBlock** — *confidential state as a load bearing wall*

07 **The agent economy** — *matches, offers and the shade graph*

08 **Adoption** — *why she stays, why brands come, why crypto disappears*

09 **Roadmap** — *the Blitz weekend and the year after*

10 **Brand, naming and legal notes** — *Aurora K, the working title, disclaimers*

Hue is a working title of Aurora K. Nothing in this document is legal, financial or investment advice. Statistics cited are drawn from published research and reporting; sources are noted where they appear.

01

THE PROBLEM

Beauty tech fails the women it studies least

Every beauty app asks a woman for her face. Almost none of them were built to understand it, and none of them let her keep it.

The evidence has been public for years. A landmark 2018 study of commercial facial analysis systems found error rates near 35 percent for darker skinned women; lighter skinned women were misread around 7 percent of the time, and lighter skinned men under 1 percent. Foundation shade tools routinely misread deep complexions because their undertone models were calibrated on lighter ones. Personal colour analysis, the twelve season system now sold as a luxury service in Seoul and Los Angeles for several hundred dollars a session, was designed decades ago around Caucasian colouring; women with deep skin are still shoved into a default season by both consultants and software. Hair recommendation engines barely comprehend type 4 curl patterns. In parts of Africa and Asia, counterfeit cosmetics and mercury laden skin lighteners remain a genuine health hazard, with no trustworthy verification layer between a woman and a jar.

Underneath all of it sits one cause: the training data is missing. Machine vision knows less about Black and brown women because the datasets contain less of them. And every serious attempt to close that gap has been extractive; faces scraped, uploaded, warehoused and monetised by companies the women never met, under terms they never read.

> *The data that would fix beauty AI is exactly the data of the women it fails. It is scarce, which makes it valuable. It is taken, which makes it theft.*

So the industry sits in a loop. It cannot serve these women without their data; it cannot be trusted with their data; so it keeps guessing, and it keeps getting them wrong. The cost lands on her: mismatched foundation, colours that never suited her, and a face print on a server she cannot audit or delete.

The gap in one sentence

There is no product on any platform where a woman of any complexion can hold a verified, portable profile of her own colouring and features that works for her, earns for her, and can never be seen by anyone she has not authorised, including the company that built the app.

02

THE IDEA

Ownership flips the data problem

Hue turns one selfie into a private, portable identity of colour and form: her undertone, her season, her face geometry, her curl pattern. We call it her Palette. She owns it the way she owns her keys, because it is secured by them.

The Palette is not stored on our servers; we do not run any that could hold it. It lives as encrypted state inside a Private Ephemeral Rollup on Solana, computed and queried inside hardware secured enclaves. Only she can decrypt it. When a brand wants to know whether its new lipstick suits her, the question travels to the enclave, the answer comes back as a plain match or no match, and her data never moves. When the answer creates value, she is paid for it, privately.

This inverts the entire arrangement of beauty tech. Today the woman is the raw material; here she is the shareholder. The dataset the industry has wanted for a decade finally gets built, but it gets built one consenting owner at a time, and it cannot be scraped, sold or subpoenaed into a spreadsheet, because no complete copy of it exists anywhere at all.

Three commitments define the product

- **Accuracy for deep skin first.** Undertone and shade logic is built and tested across the full Monk skin tone scale from day one; light skin is the edge case we also support, not the default we start from.
- **Nothing leaves without consent.** Raw images are processed on the device wherever possible. Any cloud analysis is opt in, encrypted in transit, discarded after inference, and produces only structured traits, never stored pixels.
- **Value flows to the owner.** Every query against her Palette is permissioned by her and priced for her. If her data trains something, teaches something or sells something, she is in the transaction, not under it.

> *The women the beauty industry failed become the owners of the dataset that fixes it.*

Beauty is the entry point, not the boundary. The same Palette answers questions about clothing colours, headwrap and hijab shades, eyewear shapes, hair products by curl type, and eventually sizing. Anything with a colour or a contour can be checked against it. That is why we describe Hue as a protocol wearing a consumer app, rather than an app hoping to become one.

03

THE EXPERIENCE

What she sees, taps and earns

She should never need the word blockchain. She unlocks the app with her face or fingerprint, and the one thing we can never see is that same face.

Minute one: the reading

She opens Hue on her Seeker or any Android phone and takes a short guided selfie in natural light. On the device itself, the app maps her facial geometry and samples calibrated skin regions to read her undertone. If she opts in, a vision model composes the fuller reading: face shape, feature notes, her seasonal palette, eyewear guidance, hairstyle direction. The result appears as a set of clean visual cards, short labels, no paragraphs; the analysis graphics this project began with, now produced for her rather than about her.

Minute three: the vault

The structured traits are encrypted on the device and committed to her Palette vault, a Solana account delegated into a Private Ephemeral Rollup. She signs with a passkey or the Seeker's Seed Vault; there is no seed phrase to write down and no email to surrender. A quiet line of copy tells her the truth in one breath: your reading is sealed, only you hold the key.

Every day after: the agent

Her agent works against the sealed Palette. She screenshots a dress and asks if the colour is hers; the agent checks it against her season inside the enclave and answers in under a second. Brands submit match offers: a foundation sample for cool deep undertones, a paid shade survey, an early drop for true autumns. The agent surfaces only what fits, she accepts what she likes, and small private payments in USDC arrive without any public trail announcing what her skin, her hair or her face had to do with it.

The demo line that closes the pitch

She unlocked it with her face. She earned from her colouring. And no one, including us, ever saw either.

04

ARCHITECTURE

Device, agent, enclave, chain

Hue is arranged in four layers. Sensitive raw data stays at the top; only sealed or derived state moves downward. Each arrow below is a boundary with an explicit rule about what may cross it.



The flow of one match, end to end

A brand posts an offer with target traits. Her agent forwards a match request carrying her authorisation token. Inside the enclave, the match engine decrypts her Palette, compares traits, and emits a single bit plus a signed receipt. If she accepts, the payout executes privately and a consent record commits to the base layer. Data revealed to the brand: one yes. Gas paid by her: none.

05

PRIVACY AND SECURITY

The threat model, answered

A public ledger of women's undertones, curl patterns and face geometry would not be a product; it would be a discrimination machine. Hue exists on chain only because confidential execution makes it safe to.

THREAT	HOW THE INDUSTRY FAILS TODAY	HOW HUE ANSWERS
Server breach	Face data warehoused in cloud buckets leaks in bulk.	There is no warehouse. Palettes exist only as per user encrypted state inside enclaves; a breach of our infrastructure yields nothing readable.
Chain surveillance	Public state would expose ethnic and biometric traits per wallet, forever.	Traits never touch public state. Match queries and payouts run inside the Private Ephemeral Rollup; the base layer sees receipts, not reasons.
Brand overreach	SDKs and pixels harvest far beyond stated purpose.	Permission Program ACLs are per field and revocable. A brand cleared to query undertone cannot query geometry, and every query is logged to her consent record.
Company overreach	The app maker is the biggest data buyer of all.	Aurora K holds no decryption keys, by architecture rather than policy. We cannot sell what we cannot see.
Stolen phone	Logged in apps expose everything.	Keys live in the Seed Vault or platform passkey store, gated by her biometrics, which never leave the device's secure hardware.
Model bias	Undertone models trained on light skin misread deep skin.	Calibration and evaluation across the full Monk scale is a release gate, not a roadmap item; a reading that fails deep skin does not ship.

Biometrics, stated plainly

Her face unlocks her keys; her keys sign transactions; the transactions carry signatures only. Face ID, fingerprint and Seed Vault authentication all happen inside the phone's secure hardware, and Solana's native secp256r1 verification lets those passkey signatures validate on chain directly. No email, no password, no seed phrase, and no biometric template ever transmitted anywhere. This is the difference between privacy as a setting and privacy as a structure.

06

WHY SOLANA, WHY MAGICBLOCK

Confidential state as a load bearing wall

The honest answer to "why put this on a blockchain" has always been "you should not"; until confidential execution, a chain was the worst possible home for a woman's biometric traits. That is precisely what changed.

MagicBlock's Private Ephemeral Rollups run standard Solana transactions inside Intel TDX secured enclaves: hardware verified, auditable, and integrated with fine grained access control through the Permission Program. Every primitive in the stack is load bearing for Hue rather than decorative. The private state accounts hold the Palette. The Permission Program enforces her per field consent. The enclave computes matches without revealing inputs. The Private Payments API compensates her without publishing why. Ephemeral rollup speed lets her agent run a whole shopping session of checks in real time, gasless, then settle back to the base layer.

Remove any one of those and the product collapses into either a trust us app or a surveillance ledger. That is the strongest form of the "why crypto" answer a consumer project can give: not that the chain adds a token, but that without confidential, user owned, neutrally settled state, the product is unbuildable.

What Solana gets in return

- A consumer demographic the ecosystem has barely touched: women, worldwide, arriving through beauty and fashion rather than finance.
- A flagship consumer case for Private Ephemeral Rollups beyond payments and DeFi, where privacy is ethically mandatory rather than a feature checkbox.
- A Seeker native app in a category the dApp store lacks, with passkey onboarding that never mentions crypto.
- A reusable primitive: any Solana app can query a user's Palette with consent, and none can ever see it.

> *Hue does not ask Solana users to care about beauty. It asks women who care about beauty to use Solana without noticing.*

07

THE AGENT ECONOMY

Matches, offers and the shade graph

Her data does not sit still; it works. The agent is her broker, and the enclave is the negotiating table where she never has to show her hand.

For her

The agent answers the daily questions: is this dress my colour, does this foundation match, which of these frames suits my face. It filters brand offers so only genuine fits surface. It maintains her limits: which fields may be queried, by whom, at what price, and it can be told to decline entire categories forever. Each accepted match pays her in USDC through private settlement, small amounts that add up precisely because they cost her nothing but a tap.

For brands

Brands finally get truthful demand signals for underserved shades without holding a single face. A cosmetics house can learn that four hundred verified deep cool undertone customers matched its shade gap this month; it cannot learn who they are. Sampling budgets stop being sprayed at demographics and start landing on confirmed matches. Compliance teams get hardware attestation instead of privacy policy promises.

The shade graph

The compounding asset is the cross brand shade graph: the living map of which shades correspond across brands for which undertones, a body of knowledge that today exists only as scattered forum folklore. Every consented match adds an edge. Because contributions are enclave sealed and owner permissioned, the graph can be queried in aggregate but never exported; a competitor cannot scrape it, and neither can we. The moat is not the idea. The moat is thousands of women who own their piece of the map and get paid when it is consulted.

Defensibility, honestly stated

Ideas are copyable and we will not pretend otherwise. What compounds here is consented trust, accumulated one woman at a time, in a demographic that technology has burned repeatedly. Trust is earned, not forked; and a copy of Hue without the owners of the data is a copy of an empty vault.

08

ADOPTION

Why she stays, why brands come, why crypto disappears

Three groups have to keep choosing Hue for it to matter: the woman every week, the brand every quarter, and the newcomer who never learns the word blockchain. Each has a plain reason to.

Why she uses it daily

The honest test is whether she would open the app if the earnings did not exist, and the answer has to be yes. Hue earns its place by owning purchase moments, which happen constantly: a screenshot of a dress at 11pm gets an instant verdict against her palette; two foundations held up in a store get a shelf scan and a match; a scarf against a blouse gets a one second answer from the camera. Colour decisions are a weekly anxiety currently served by guesswork, a friend on WhatsApp and forum threads; the agent replaces all three with an answer calibrated to her actual skin. Around that core sit the softer hooks: palette cards designed to be shared, since declaring a season is already a thing women post; seasonal wardrobe audits; and earnings that accumulate quietly in the corner, a reward she discovers rather than a pitch she was sold. Payouts are the retention bonus, never the acquisition promise; products that lead with get paid attract mercenaries and repel everyone else.

Why brands come here first

Because the alternative does not exist. No broker, panel or ad platform sells verified, consented, undertone level demand; the incumbents can target a demographic, not a shade. Hue can tell a cosmetics house that four hundred verified deep cool customers matched its shade gap this month and sixty accepted samples, which is not an insight, it is a purchase order. Three incentives compound: sampling budgets land on confirmed matches instead of demographic guesses; foundation mismatch returns attack margin and Hue attacks their cause; and, quietly the largest, liability. Under GDPR, Illinois BIPA and the laws following them, holding biometric data is a lawsuit generator, and Hue gives brands face level matching while never touching a face, with hardware attestation for the compliance file. The sequence is deliberate: indie and inclusive first brands sign first, because they are starving for exactly this data and move fast; their case studies become the lever that pulls the majors, who never move first and always follow proof.

Why crypto disappears

Crypto vocabulary appears exactly once in the app, in the settings fine print for the curious. Everywhere else, translation. She does not create a wallet; she signs in with her fingerprint. She does not hold USDC; she has earnings shown in her currency, cashable or spendable on offers. She does not pay gas; nothing is ever paid, because session keys make every action free and silent. Losing her phone does not lose her Palette, because recovery through synced passkeys and encrypted backup is built from day one. The sentence she is given is one she will actually like: your data is so private that even we cannot see it. The continuous incentive is simply that the product keeps being right about her; it saves money weekly, it works for her skin where nothing else did, and value accrues while she does nothing. If she ever discovers it runs on Solana, the designed reaction is mild curiosity, not alarm; and that is how a new demographic is onboarded to a chain, by never being asked to care about it.

09

ROADMAP

The Blitz weekend and the year after

Hackathon scope: one selfie, one sealed Palette, one paid match

Day one delivers the reading and the vault: camera flow with lighting guidance, on device landmarking and undertone sampling, the opt in vision reading rendered as visual cards, client side encryption, and the Anchor program with delegation into a devnet Private Ephemeral Rollup. Sign in is passkey or Seed Vault from the first screen; no email field exists anywhere in the app.

Day two delivers the agent and the money: a mock brand posts offers, the match engine runs inside the delegated state, the agent surfaces one true fit, she accepts, and a private USDC payout lands with a consent receipt committed on chain. The demo ends on a single screen: three matches, a small balance earned, and the sentence that carries the project, your face never left this phone.

Everything else is deliberately cut: no try on rendering, no live brand integrations, no token, no social features. A complete loop beats a broad sketch.

After the weekend

- **Quarter one:** Seeker dApp store submission; wardrobe and clothing colour checks; shade graph v1 across ten launch brands; audits of the vault and match programs.
- **Quarter two:** hair and curl type module; headwrap and modest fashion palettes; consultation marketplace where verified colourists read Palettes inside the enclave under the same consent rules.
- **Quarter three:** the Palette as an open primitive: a consent gated query interface any Solana consumer app can integrate, with revenue sharing to owners; counterfeit cosmetics verification pilot in two Global South markets.



THE HUE SYSTEM PALETTE · INK 2A1C4E · PLUM 3A2358 · AURORA 6C4BD8 · TEAL 1E5F5B · GOLD C89B4B · COPPER B4633A · CREAM F5EFE4

10

BRAND, NAMING AND LEGAL NOTES

Aurora K, the working title, and what remains to verify

Brand. Hue is developed under Aurora K. The product name in this document is a working title chosen for the pitch: one word, one syllable, and literally the subject matter, since a hue is what the entire system reads, seals and serves. Whether it survives to launch depends on clearance work described below.

Naming clearance, stated candidly. Short common words carry crowded trademark registers. "Hue" in particular coexists with well known marks in adjacent categories, including consumer electronics and lighting, and short marks in cosmetics classes are common. Before public launch the team should run searches across the relevant registers (USPTO, EUIPO, WIPO Global Brand Database and the local register for any launch market), check app store and domain availability, and take advice from a qualified trademark attorney; nothing in this paper is legal advice and no clearance has been performed. Candidate alternatives worth screening in the same pass: Huetone, Undertone, Aurine, Kolora, and the plain house brand Aurora K itself.

Biometric and data law. The architecture is designed so that biometric templates never leave device secure hardware and raw images are not retained, which is the correct posture under regimes such as GDPR, Illinois BIPA and comparable laws; formal review is still required before any mainnet launch handling real users, and consent language in the app must match what the system actually does. Because it does, that review should be pleasant for once.

Financial surface. Payouts are stablecoin transfers for consented data queries; the project has no token and this document offers nothing for sale. Compliance obligations for payments vary by jurisdiction and must be reviewed before mainnet payouts are enabled. MagicBlock's Private Ephemeral Rollups enforce jurisdictional compliance at ingress, which helps, and does not substitute for counsel.

Attribution of claims. The facial analysis error figures cited in section 01 refer to the Gender Shades study (Buolamwini and Gebru, 2018). Descriptions of MagicBlock systems reflect their public documentation as of July 2026. The Monk skin tone scale referenced throughout is the ten shade scale developed by Dr. Ellis Monk.

HUE · a working title of Aurora K · Whitepaper v0.1 · July 2026 · Confidential draft, prepared for the MagicBlock Solana Blitz hackathon. Written by the founding team. This document will change; the commitments in section 02 will not.